

"A mind stretched to new measurements never returns to its original dimensions."

CONTACT

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PROFILE

Computer Science student and aspiring software engineer focused on agentic AI systems and architecture, with hands-on project experience in C++, Java, and Python, complemented by years of professional IT support and client-facing experience.

EDUCATION

BSc. Computer Science & Information Technology
University of KwaZulu-Natal | ***Jan 2023 – Nov 2026 · In Progress***

Double major in Computer Science and Mathematics, taking additional mathematics coursework beyond the standard single-major track for a stronger quantitative and analytical foundation.

National Senior Certificate, Technical Stream
George Campbell School of Technology | *Jan 2017 – Nov 2020*

Electrical Technology (Digital Systems): Circuit Analysis, logic Systems, Communications, Amplifiers, RLC circuits, and Protective Devices.

Hands-on with PCB layout/design basics, soldering, microcontroller-based circuit building, and electrical measurement and troubleshooting.

TECHNICAL SKILLS

- Agentic AI Architecture & Design
- LLM Tool Use & Evaluation
- C++ (OOP, multi-threading)
- Java & Swing (JDBC)
- Python (ML / NLP / LLMs)
- SQL & Database Design
- Qt6 GUI Development
- Git & Version Control
- IT Hardware & CRM / Accounts
- Android Development (Kotlin)

LANGUAGES

· English (Home Language)
· Afrikaans (First Additional)
· isiZulu (Beginner)

REFERENCES

Mr Reece+ Sher – RM Graphics Solutions (Owner & CEO)
info@rmgraphics.co.za / Cell Number on Request

JAWAD KACHBAL

SOFTWARE ENGINEER • IT SUPPORT TECHNICIAN • COMPUTER SCIENCE STUDENT
AGENTIC AI SYSTEMS • AI ARCHITECTURE & DESIGN • LLM TOOL USE & EVALUATION

PROJECTS *For more projects and details, visit my website and GitHub*

Confident Agent — Agentic AI System for Detecting LLM API Hallucinations

Agentic AI · System Architecture · LLM Tool Use | ***COMP301 · Team of 5 · Group Leader*** *Github Link*

Agentic AI system that stops hallucinated REST API calls in LLM-generated code from reaching the user: a local LLM (Qwen3.5-9B) paired with an in-loop static verification tool.

- Architected the agent end to end: an in-loop, function-calling verifier (six-layer check against frozen OpenAPI specs) feeding a confidence-scored deliver / flag / correct / withhold policy
- Designed a 1,000-task benchmark and independent sandboxed evaluator; ran paired baseline-vs-agent experiments across three quantizations with a pre-registered analysis plan
- Hallucinated code reaching the user fell from 50.8–55.2% to 1.6–2.3% (correct code up to 50.4–51.5%)

Nexus Logistics — Inventory Management System

C++ · CLI + Qt6 GUI (bonus) | ***COMP315 Group Project · Team of 7*** *Github Link*

Multi-threaded inventory and order-management system for a simulated 5-warehouse logistics network, with an object-oriented product hierarchy and CSV-backed persistence.

- Designed a polymorphic OOP hierarchy (Standard, Discounted, Taxable products) with multi-threaded order processing (Producer/Consumer Relationship) and per-warehouse analytics via a custom AnalyticsManager
- Built a bonus Qt6 GUI (dashboard, inventory, analytics, orders) on top of the CLI backend via an adapter layer

Starlight Carnival Management System

Github Link

Java, Swing, MySQL/JDBC | ***COMP306 Group Project***

Full-stack desktop database application for managing a carnival's attractions, staff, and access passes.

- Modelled 3 disjoint subtype hierarchies with role-based views, stored procedures, and triggers; built a Java Swing CRUD/reporting UI over MySQL via JDBC
- Implemented role-based login (Admin, Sales, Maintenance, Guest) with distinct MySQL accounts

Music Square — Mood-Based Music Player + PixelPlayer Android App Integration

Github Link

Python (Flask, librosa), JavaScript/HTML/CSS | ***Personal Project · Solo***

Local player that analyses FLAC audio directly (no tags/cloud lookups), plotting tracks on a 2D mood square.

- Extracted tempo, loudness, onset density, brightness, and tonality with librosa, combining library-wide z-scores into weighted arousal/valence mood axes
- Built a Flask backend for parallelized analysis/streaming and a vanilla JS/Canvas frontend with zoom/pan, mood-based playlist building, and OS media-key integration
- Currently working on Porting the Analysis Algorithm to Android via the Open Source PixelPlayer Music Player

NLP Sentiment Classification — Naive Bayes vs LSTM vs BERT

Python, scikit-learn, PyTorch/Transformers | ***COMP316 Group Project · Team of 3*** ***Github Link***

- Built and curated a 60,000-review Steam game dataset via the Steam Web API for sentiment analysis
- Designed and implemented the BERT fine-tuning pipeline, achieving 90.3% accuracy and a 90.3% F1-score, outperforming the Naive Bayes and Bi-LSTM baselines (≈84%) in sentiment classification

EXPERIENCE

IT Support & Hardware Technician

RM Graphic Solutions | ***Jan 2021 – Present · 5 yrs 7 mo · Part-time | Durban, KwaZulu-Natal***

Provide end-to-end IT support across desktops and laptops: diagnosing hardware and software faults, building custom PCs, and handling installations, upgrades, backups, and data recovery.

- Deliver technical support and preventative maintenance to minimise downtime
- Troubleshoot and repair hardware and software faults across a range of systems

Sales Representative (Junior --> Senior)

RM Graphic Solutions | ***Jan 2023 – Jan 2026 · 3 yrs 1 mo · Part-time | Durban, KwaZulu-Natal***

Progressed from Junior to Senior Sales Representative, taking on increasing responsibility in sales, account management, and print shop operations, managing projects from enquiry through to delivery.

- Ran day-to-day print shop operations, coordinating workflow between sales and production
- Developed new business opportunities and managed key client accounts.
- Consulted walk-in, phone, email, and WhatsApp clients on printing and branding solutions
- Prepared quotations, negotiated pricing, processed orders, and managed invoicing.
- Oversaw projects from concept to production, ensuring quality standards and on-time delivery.